

Binu Prajapati

Computer Engineering Final-Year Student | AI/ML and Data Science | Backend and FullStack Developer

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Professional Summary

Final-year Computer Engineering student at Khwopa Engineering College with hands-on experience in backend and full-stack development, data pipelines, and machine learning applications across academic and hackathon projects. Practical background building Django-based backends, REST APIs, and database-driven applications using PostgreSQL, MongoDB, MySQL, and SQLite, along with designing multi-stage pipelines for text, audio, and video data. Experienced in Python-based data processing with Pandas and NumPy and integrating ML/NLP components into end-to-end applications. Strong Python fundamentals with a fast-learning mindset and experience working in fast-paced team environments.

Technical Skills

Programming Languages	Python, C, C++
AI/ML & Data Science	NLP, LLM APIs, PyTorch, Streamlit, Hugging Face, Pandas, NumPy, scikit-learn
Backend and Web Development	Django, FastAPI, Flask, REST APIs, HTML, CSS, JavaScript
Data Engineering	Data Pipelines, Data Cleaning and Preprocessing
Databases	MySQL, PostgreSQL, SQLite, MongoDB
Tools & Technologies	Git, VS Code, Jupyter Notebook, Google Colab, Postman, L ^A T _E X

Projects

RECONNECT: Multi-Agent Generative AI Cognitive Memory Assistance in Dementia Care System(Alzheimer's disease) *Final-Year Major Team Project*

Tech Stack: Python, PyTorch, Whisper, PyAnnote, ECAPA-TDNN, InsightFace, TalkNet-ASD, SQLite, FastAPI, AI/ML/
View Project | [View Project](#)

- Developed a memory assistance system for dementia patients that continuously processes speech and visual information to identify people, understand conversations and store contextual memories.
- Implemented a real-time audio perception pipeline integrating voice detection, Whisper-based speech-to-text, PyAnnote speaker diarization, and ECAPA-TDNN speaker recognition for continuous speaker-aware transcription.
- Designed identity verification pipeline combining InsightFace-based face recognition with TalkNet active speaker detection to associate unknown voices with corresponding faces and support dynamic identity enrollment.
- Built a memory processing architecture that extracts entities and contextual events enabling structured storage and semantic retrieval of relevant past experiences.

POLARISAI: Controversy Detection & AI Prevention System *Hackathon Team Project*

Tech Stack: Python, FastAPI, NLP, LLM APIs, Reddit API, Browser Extension, AI/ML | View Project | [View Project](#)

- Developed an AI-powered controversy intelligence platform and browser extension enabling users to analyze claims and online discussions directly from web content.
- Implemented an explainable controversy scoring framework by engineering a multi-stage NLP pipeline integrating claim extraction, Reddit multi-thread retrieval, stance detection, emotion analysis, topic sensitivity assessment, and audience divergence analysis.
- Designed a recommendation component that generated less controversial alternative statements and constructive reformulations to encourage balanced communication and reduce inflammatory phrasing.
- Collaborated within a fast-paced hackathon team environment to rapidly prototype, develop and present the solution, securing **1st Runner-Up position at NationalAI Hackathon 2026**.

Logic Lens: Fallacy Detector & AI Reasoning Assistant *Hackathon Team Project*

Tech Stack: Python, AI/ML, NLP, LLM APIs, Browser Extension | [View Project](#)

- Developed an AI-powered logical fallacy detection and reasoning assistant delivered through a browser extension, enabling users to analyse online content and arguments directly within their browsing experience.
- Integrated Natural Language Processing and Large Language Model APIs to identify fallacious reasoning patterns, evaluate argumentative structures, and generate contextual explanations with reasoning-based feedback.

- Designed a lightweight and user-friendly extension workflow for real-time text analysis, critical thinking assistance, and educational feedback across web platforms.
- Collaborated within a fast-paced hackathon team environment to rapidly prototype, develop, and present the solution, leading to **winning the AI/ML Track at KU Hackathon 2025**.

RentEra: Peer-to-Peer Rental & Management System

Third-Year Major Team Project

Tech Stack: Django, Python, SQLite, JavaScript, AI/NLP (Chatbot), eSewa & PayPal APIs | [View Project](#)

- Developed a Django-based peer-to-peer rental platform with secure authentication and citizenship and land ownership verification to improve trust between renters and owners.
- Engineered an intelligent recommendation engine that dynamically suggests properties based on user navigation patterns, clickstream data, and precise geolocation determination.
- Integrated an AI-powered chatbot and real-time communication system to automate renter-owner interactions, alongside a custom automated expiry logic for property listings to maintain data freshness.
- Built a robust backend for financial transactions, successfully integrating eSewa and PayPal gateways to facilitate secure rental payments and tracking.

Painting Muse- eCommerce website

Individual Project

Tech Stack: HTML, CSS, JS, python, postgres, Django | [View Project](#)

- Developed a full-stack e-commerce platform for art commerce featuring multi-criteria filtering, secure PayPal integration, and a category-based search system to streamline the buying and selling of paintings.

Bus Ticket Reservation System

Team Project

Tech Stack: C++, graphics.h | [View Project](#)

- Developed a GUI-based desktop application for bus reservation featuring an interactive seat layout, automated ticket booking, and route management using C++ graphics libraries.

Additional Mini Projects: Cloudfusionn | Chatbot- UFONepal | Pocketmedid | Echo| Tabula| Photobooth| Wordpress webiste (Anime Store) | Task manager | Graphical linkedlist representator

Achievements & Certifications

- **Winner** — 1st Runneer-Up at NationalAI HAckathon 2026
- **Winner** — AI/ML Track, KU Hackathon 2025
- 10-day AI & ML workshop by ITNSP
- 7-day Jira Workshop by Delta Dynamics Academy
- KhwopaCeel 2026 participant
- Postman training

Leadership & Extracurricular

- **Workshop Lead** — Nobel Internship: Facilitated a 4-day WordPress web design workshop, guiding participants through hands-on exercises in site development and design.
- Actively participated in hackathons and competitive technical challenges, demonstrating problem-solving under real-world constraints.
- Experienced in Agile-style team-based software development with roles spanning UI/UX, backend, and project coordination.

Education

Bachelor of Engineering in Computer Engineering

2022 – Present

Khwopa Engineering College, Bhaktapur | Purbanchal University

Higher Secondary Education (+2 Science)

2022

Khwopa Higher Secondary School, Bhaktapur

Secondary Education Examination (SEE)

2020

Bagiswori Secondary School, Bhaktapur